DBMS Lab Assignment 9 - Solution

Retrieving data from multiple tables (joins). Group A uses Customer, Movie and Invoice; Group B uses the EMP table. Each question shows the Oracle SQL followed by its output.

*Note: outputs were produced by running equivalent statements on a SQLite database with the same tables and data, formatted like SQL*Plus.*

Create the tables

SQL

```
CREATE TABLE Customer (
  Cust_id VARCHAR2(5)  CONSTRAINT pk_customer PRIMARY KEY
                       CONSTRAINT chk_cust_id CHECK (Cust_id LIKE 'A%'),
  Fname   VARCHAR2(20) CONSTRAINT nn_cust_fname NOT NULL,
  Lname   VARCHAR2(20),
  Area    VARCHAR2(2)  CONSTRAINT nn_cust_area NOT NULL,
  Phone   NUMBER(10));

CREATE TABLE Movie (
  Mv_no  NUMBER(3)    CONSTRAINT pk_movie PRIMARY KEY,
  Title  VARCHAR2(30) CONSTRAINT nn_mv_title NOT NULL,
  Type   VARCHAR2(15) CONSTRAINT nn_mv_type  NOT NULL,
  Star   VARCHAR2(5)  CONSTRAINT nn_mv_star  NOT NULL,
  Price  NUMBER(6)    CONSTRAINT nn_mv_price NOT NULL);

CREATE TABLE Invoice (
  Inv_no      VARCHAR2(5) CONSTRAINT pk_invoice PRIMARY KEY
                          CONSTRAINT chk_inv_no CHECK (Inv_no LIKE 'I%'),
  Mv_no       NUMBER(3)   CONSTRAINT fk_inv_movie REFERENCES Movie(Mv_no),
  Cust_id     VARCHAR2(5) CONSTRAINT fk_inv_cust  REFERENCES Customer(Cust_id),
  Issue_date  DATE        CONSTRAINT nn_inv_issue  NOT NULL,
  Return_date DATE        CONSTRAINT nn_inv_return NOT NULL);
-- Customer and Movie must exist before Invoice, which references both.
```

OUTPUT

```
Table created.

Table created.

Table created.
```

Note: Constraints are read from the sheet column by column: Customer = PK on Cust_id, NOT NULL on Fname and Area; Movie = PK on Mv_no, NOT NULL on the other four; Invoice = PK, two FKs, NOT NULL on both dates. The '__' entries in the data are NULLs (Ravi's phone, Rukmini's last name).

Insert the data

SQL

```
-- Customer
INSERT INTO Customer VALUES ('A01','Ivan','Bayross','SA',6125467);
INSERT INTO Customer VALUES ('A02','Vandana','Saiteal','MU',5560379);
INSERT INTO Customer VALUES ('A03','Pramada','Jauguste','DA',4560389);
INSERT INTO Customer VALUES ('A04','Basu','Navindi','BA',6125401);
INSERT INTO Customer VALUES ('A05','Ravi','Shreedhar','NA',NULL);
INSERT INTO Customer VALUES ('A06','Rukmini',NULL,'GH',5125274);

-- Movie
INSERT INTO Movie VALUES (1,'Bloody','Action','JC',181);
INSERT INTO Movie VALUES (2,'The Firm','Thriller','TC',200);
INSERT INTO Movie VALUES (3,'Pretty Woman','Romance','RG',151);
INSERT INTO Movie VALUES (4,'Home Alone','Comedy','MC',150);
INSERT INTO Movie VALUES (5,'The Fugitive','Thriller','MF',200);
INSERT INTO Movie VALUES (6,'Coma','Suspense','MD',100);
INSERT INTO Movie VALUES (7,'Dracula','Horror','GO',150);
INSERT INTO Movie VALUES (8,'Quick Change','Comedy','BM',100);
INSERT INTO Movie VALUES (9,'Gone with the Wind','Action','CB',200);
INSERT INTO Movie VALUES (10,'Carry on Doctor','Comedy','LP',100);

-- Invoice
INSERT INTO Invoice VALUES ('I01',4,'A01',DATE '1993-07-23',DATE '1993-07-25');
INSERT INTO Invoice VALUES ('I02',3,'A02',DATE '1993-08-12',DATE '1993-08-15');
INSERT INTO Invoice VALUES ('I03',1,'A02',DATE '1993-08-15',DATE '1993-08-18');
INSERT INTO Invoice VALUES ('I04',6,'A03',DATE '1993-09-10',DATE '1993-09-13');
INSERT INTO Invoice VALUES ('I05',7,'A04',DATE '1993-08-05',DATE '1993-08-08');
INSERT INTO Invoice VALUES ('I06',2,'A06',DATE '1993-09-18',DATE '1993-09-21');
INSERT INTO Invoice VALUES ('I07',9,'A05',DATE '1993-07-07',DATE '1993-07-10');
INSERT INTO Invoice VALUES ('I08',9,'A01',DATE '1993-08-11',DATE '1993-08-14');
INSERT INTO Invoice VALUES ('I09',5,'A03',DATE '1993-07-06',DATE '1993-07-09');
INSERT INTO Invoice VALUES ('I10',8,'A06',DATE '1993-09-03',DATE '1993-09-06');
COMMIT;
```

OUTPUT

```
6 rows created. (Customer)
10 rows created. (Movie)
10 rows created. (Invoice)

CUST_ID  FNAME    LNAME      AREA    PHONE
-------  -------  ---------  ----  -------
A01      Ivan     Bayross    SA    6125467
A02      Vandana  Saiteal    MU    5560379
A03      Pramada  Jauguste   DA    4560389
A04      Basu     Navindi    BA    6125401
A05      Ravi     Shreedhar  NA
A06      Rukmini             GH    5125274

6 rows selected.

MV_NO  TITLE               TYPE      STAR  PRICE
-----  ------------------  --------  ----  -----
    1  Bloody              Action    JC      181
    2  The Firm            Thriller  TC      200
    3  Pretty Woman        Romance   RG      151
    4  Home Alone          Comedy    MC      150
    5  The Fugitive        Thriller  MF      200
    6  Coma                Suspense  MD      100
    7  Dracula             Horror    GO      150
    8  Quick Change        Comedy    BM      100
    9  Gone with the Wind  Action    CB      200
   10  Carry on Doctor     Comedy    LP      100

10 rows selected.

INV_NO  MV_NO  CUST_ID  ISSUE_DATE  RETURN_DATE
------  -----  -------  ----------  -----------
I01         4  A01      23-JUL-93   25-JUL-93
I02         3  A02      12-AUG-93   15-AUG-93
I03         1  A02      15-AUG-93   18-AUG-93
I04         6  A03      10-SEP-93   13-SEP-93
I05         7  A04      05-AUG-93   08-AUG-93
I06         2  A06      18-SEP-93   21-SEP-93
I07         9  A05      07-JUL-93   10-JUL-93
I08         9  A01      11-AUG-93   14-AUG-93
I09         5  A03      06-JUL-93   09-JUL-93
I10         8  A06      03-SEP-93   06-SEP-93

10 rows selected.
```

GROUP A - 1. Customer names and movie number of all customers who have been issued a movie

SQL

```
SELECT c.Fname, c.Lname, i.Mv_no
FROM   Customer c JOIN Invoice i ON c.Cust_id = i.Cust_id;
```

OUTPUT

```
FNAME    LNAME      MV_NO
-------  ---------  -----
Ivan     Bayross        4
Vandana  Saiteal        3
Vandana  Saiteal        1
Pramada  Jauguste       6
Basu     Navindi        7
Rukmini                 2
Ravi     Shreedhar      9
Ivan     Bayross        9
Pramada  Jauguste       5
Rukmini                 8

10 rows selected.
```

2. Movie number and title issued to Ivan

SQL

```
SELECT m.Mv_no, m.Title
FROM   Customer c JOIN Invoice i ON c.Cust_id = i.Cust_id
                  JOIN Movie m   ON i.Mv_no   = m.Mv_no
WHERE  c.Fname = 'Ivan';
```

OUTPUT

```
MV_NO  TITLE
-----  ------------------
    4  Home Alone
    9  Gone with the Wind

2 rows selected.
```

3. Title, customer id and movie number for all movies that are issued

SQL

```
SELECT m.Title, i.Cust_id, m.Mv_no
FROM   Movie m JOIN Invoice i ON m.Mv_no = i.Mv_no;
```

OUTPUT

```
TITLE               CUST_ID  MV_NO
------------------  -------  -----
Home Alone          A01          4
Pretty Woman        A02          3
Bloody              A02          1
Coma                A03          6
Dracula             A04          7
The Firm            A06          2
Gone with the Wind  A05          9
Gone with the Wind  A01          9
The Fugitive        A03          5
Quick Change        A06          8

10 rows selected.
```

4. Title and type of the movies issued to Vandana

SQL

```
SELECT m.Title, m.Type
FROM   Customer c JOIN Invoice i ON c.Cust_id = i.Cust_id
                  JOIN Movie m   ON i.Mv_no   = m.Mv_no
WHERE  c.Fname = 'Vandana';
```

OUTPUT

```
TITLE         TYPE
------------  -------
Pretty Woman  Romance
Bloody        Action

2 rows selected.
```

5. Names of customers who have been issued a movie of type Action

SQL

```
SELECT c.Fname, c.Lname, m.Title
FROM   Customer c JOIN Invoice i ON c.Cust_id = i.Cust_id
                  JOIN Movie m   ON i.Mv_no   = m.Mv_no
WHERE  m.Type = 'Action';
```

OUTPUT

```
FNAME    LNAME      TITLE
-------  ---------  ------------------
Vandana  Saiteal    Bloody
Ravi     Shreedhar  Gone with the Wind
Ivan     Bayross    Gone with the Wind

3 rows selected.
```

6. 'The movie taken by {Fname} {Lname} is {Title}.' for movie number >= 3

SQL

```
SELECT 'The movie taken by ' || c.Fname || ' ' || c.Lname ||
       ' is ' || m.Title || '.' AS statement
FROM   Customer c JOIN Invoice i ON c.Cust_id = i.Cust_id
                  JOIN Movie m   ON i.Mv_no   = m.Mv_no
WHERE  m.Mv_no >= 3;
```

OUTPUT

```
STATEMENT
-----------------------------------------------------
The movie taken by Ivan Bayross is Home Alone.
The movie taken by Vandana Saiteal is Pretty Woman.
The movie taken by Pramada Jauguste is Coma.
The movie taken by Basu Navindi is Dracula.
The movie taken by Ravi Shreedhar is Gone with the Wind.
The movie taken by Ivan Bayross is Gone with the Wind.
The movie taken by Pramada Jauguste is The Fugitive.
The movie taken by Rukmini  is Quick Change.

8 rows selected.
```

Note: Rukmini has no last name (NULL), and Oracle treats NULL as empty text in ||, so her line has two spaces after the name. Use NVL(c.Lname,'') or TRIM to tidy it if you prefer.

7. Customer who has been issued the movie on invoice 'I10'

SQL

```
SELECT c.Cust_id, c.Fname, c.Lname
FROM   Customer c JOIN Invoice i ON c.Cust_id = i.Cust_id
WHERE  i.Inv_no = 'I10';
```

OUTPUT

```
CUST_ID  FNAME    LNAME
-------  -------  -----
A06      Rukmini

1 row selected.
```

8. Customer name, phone and area for movies issued before August

SQL

```
SELECT c.Fname, c.Lname, c.Phone, c.Area, i.Issue_date
FROM   Customer c JOIN Invoice i ON c.Cust_id = i.Cust_id
WHERE  EXTRACT(MONTH FROM i.Issue_date) < 8;
```

OUTPUT

```
FNAME    LNAME        PHONE  AREA  ISSUE_DATE
-------  ---------  -------  ----  ----------
Pramada  Jauguste   4560389  DA    06-JUL-93
Ravi     Shreedhar           NA    07-JUL-93
Ivan     Bayross    6125467  SA    23-JUL-93

3 rows selected.
```

Note: Ravi's phone is NULL, so it shows blank. The issue date is shown only to prove the filter.

9. Movie number and title issued to Ivan, Vandana and Ravi

SQL

```
SELECT c.Fname, m.Mv_no, m.Title
FROM   Customer c JOIN Invoice i ON c.Cust_id = i.Cust_id
                  JOIN Movie m   ON i.Mv_no   = m.Mv_no
WHERE  c.Fname IN ('Ivan','Vandana','Ravi');
```

OUTPUT

```
FNAME    MV_NO  TITLE
-------  -----  ------------------
Ivan         4  Home Alone
Ivan         9  Gone with the Wind
Vandana      3  Pretty Woman
Vandana      1  Bloody
Ravi         9  Gone with the Wind

5 rows selected.
```

10. Movie number and movie name issued to customers

SQL

```
SELECT DISTINCT m.Mv_no, m.Title
FROM   Movie m JOIN Invoice i ON m.Mv_no = i.Mv_no
ORDER  BY m.Mv_no;
```

OUTPUT

```
MV_NO  TITLE
-----  ------------------
    1  Bloody
    2  The Firm
    3  Pretty Woman
    4  Home Alone
    5  The Fugitive
    6  Coma
    7  Dracula
    8  Quick Change
    9  Gone with the Wind

9 rows selected.
```

Note: DISTINCT is used because movie 9 was issued twice.

11. Type and movie numbers of movies issued to customers 'A01' and 'A02'

SQL

```
SELECT i.Cust_id, m.Mv_no, m.Type
FROM   Movie m JOIN Invoice i ON m.Mv_no = i.Mv_no
WHERE  i.Cust_id IN ('A01','A02');
```

OUTPUT

```
CUST_ID  MV_NO  TYPE
-------  -----  -------
A01          4  Comedy
A01          9  Action
A02          3  Romance
A02          1  Action

4 rows selected.
```

12. Has the movie starring Tom Cruise been issued? Print the customer id

SQL

```
SELECT i.Cust_id, m.Title, m.Star
FROM   Movie m JOIN Invoice i ON m.Mv_no = i.Mv_no
WHERE  m.Star = 'TC';
```

OUTPUT

```
CUST_ID  TITLE     STAR
-------  --------  ----
A06      The Firm  TC

1 row selected.
```

Note: The Star column holds initials, not full names, so 'Tom Cruise' is matched as 'TC' (The Firm). Yes, it was issued, to A06.

13. First and last name of customers who have been issued a movie

SQL

```
SELECT DISTINCT c.Fname, c.Lname
FROM   Customer c JOIN Invoice i ON c.Cust_id = i.Cust_id;
```

OUTPUT

```
FNAME    LNAME
-------  ---------
Basu     Navindi
Ivan     Bayross
Pramada  Jauguste
Ravi     Shreedhar
Rukmini
Vandana  Saiteal

6 rows selected.
```

Note: All six customers have at least one invoice.

14. Movie number that has never been issued to any customer

SQL

```
SELECT Mv_no, Title
FROM   Movie
WHERE  Mv_no NOT IN (SELECT Mv_no FROM Invoice);
```

OUTPUT

```
MV_NO  TITLE
-----  ---------------
   10  Carry on Doctor

1 row selected.
```

GROUP B - EMP table used for the self-joins

SQL

```
SELECT empno, ename, job, mgr, deptno FROM emp;
```

OUTPUT

```
EMPNO  ENAME   JOB         MGR  DEPTNO
-----  ------  ---------  ----  ------
 7369  SMITH   CLERK      7902      20
 7499  ALLEN   SALESMAN   7698      30
 7521  WARD    SALESMAN   7698      30
 7566  JONES   MANAGER    7839      20
 7654  MARTIN  SALESMAN   7698      30
 7698  BLAKE   MANAGER    7839      30
 7782  CLARK   MANAGER    7839      10
 7788  SCOTT   ANALYST    7566      20
 7839  KING    PRESIDENT             10
 7844  TURNER  SALESMAN   7698      30
 7876  ADAMS   CLERK      7788      20
 7900  JAMES   CLERK      7698      30
 7902  FORD    ANALYST    7566      20
 7934  MILLER  CLERK      7782      10

14 rows selected.
```

Note: The classic Oracle EMP table (SCOTT schema) was recreated here with its standard 14 rows.

B1. Employee id, name, manager id and manager's name

SQL

```
SELECT e.empno AS employee_id, e.ename AS employee_name,
       e.mgr AS manager_id, m.ename AS manager_name
FROM   emp e LEFT JOIN emp m ON e.mgr = m.empno;
```

OUTPUT

```
EMPLOYEE_ID  EMPLOYEE_NAME  MANAGER_ID  MANAGER_NAME
-----------  -------------  ----------  ------------
       7369  SMITH                7902  FORD
       7499  ALLEN                7698  BLAKE
       7521  WARD                 7698  BLAKE
       7566  JONES                7839  KING
       7654  MARTIN               7698  BLAKE
       7698  BLAKE                7839  KING
       7782  CLARK                7839  KING
       7788  SCOTT                7566  JONES
       7839  KING
       7844  TURNER               7698  BLAKE
       7876  ADAMS                7788  SCOTT
       7900  JAMES                7698  BLAKE
       7902  FORD                 7566  JONES
       7934  MILLER               7782  CLARK

14 rows selected.
```

Note: LEFT JOIN is used so that KING (the president, with no manager) still appears; a plain inner join would drop him.

B2. Employee name, manager id, manager's name and manager's manager name

SQL

```
SELECT e.ename AS employee_name, e.mgr AS manager_id,
       m.ename AS manager_name, g.ename AS managers_manager_name
FROM   emp e LEFT JOIN emp m ON e.mgr = m.empno
             LEFT JOIN emp g ON m.mgr = g.empno;
```

OUTPUT

```
EMPLOYEE_NAME  MANAGER_ID  MANAGER_NAME  MANAGERS_MANAGER_NAME
-------------  ----------  ------------  --------------------
SMITH                7902  FORD          JONES
ALLEN                7698  BLAKE         KING
WARD                 7698  BLAKE         KING
JONES                7839  KING
MARTIN               7698  BLAKE         KING
BLAKE                7839  KING
CLARK                7839  KING
SCOTT                7566  JONES         KING
KING
TURNER               7698  BLAKE         KING
ADAMS                7788  SCOTT         JONES
JAMES                7698  BLAKE         KING
FORD                 7566  JONES         KING
MILLER               7782  CLARK         KING

14 rows selected.
```

Note: Blank cells are NULLs: KING has no manager, and employees managed directly by KING have no manager's manager.